

VISHAKHAPATNAM CWC, India

WMO: 431500

Lat: 17.7217N

Lon: 83.3292E

Elev: 217

StdP: 14.58

Time Zone: 5.50 (IND)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	68.1	69.4	51.7	57.5	77.1	55.2	65.2	76.3	16.3	75.2	14.3	76.1	2.9	0	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.5	93.1	80.9	91.7	81.6	90.8	81.5	84.7	90.1	84.0	89.5	83.3	88.9	6.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
83.3	175.5	89.1	82.5	170.7	88.6	81.7	166.2	88.0	49.5	90.3	48.6	89.6	47.6	88.8	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.6	14.1	12.1	64.6	99.6	2.7	2.7	62.7	101.5	61.1	103.1	59.6	104.6	57.7	106.5		
			56.9	85.8	2.2	0.4	55.3	86.1	54.1	86.3	52.8	86.6	51.2	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	75.9	78.6	82.3	84.7	87.0	86.0	84.0	83.6	83.7	83.1	80.1	76.8
	DBStd	4.00	1.78	1.95	1.66	1.75	2.21	2.95	2.63	2.09	1.95	2.07	2.04	2.11
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11739	803	800	1001	1041	1146	1080	1053	1041	1012	1026	904	832
	CDD65	6264	338	380	536	591	681	630	588	576	562	561	454	367
	CDH74	71998	1910	3151	6161	7755	9699	8609	7399	7083	6921	6674	4271	2365
	CDH80	28482	199	642	2058	3502	5264	4352	3045	2755	2731	2491	1091	352
Wind	WSAvg	5.1	3.9	4.0	5.2	6.6	6.1	5.8	5.2	4.8	3.9	4.8	5.8	4.9
Precipitation	PrecAvg	39.2	0.4	0.5	0.5	1.2	1.9	4.1	5.2	6.1	6.8	8.9	3.4	0.3
	PrecMax	57.1	3.5	4.5	7.6	10.9	8.3	13.4	14.0	16.9	15.9	26.3	14.4	3.0
	PrecMin	25.2	0.0	0.0	0.0	0.0	0.2	0.7	1.1	1.1	1.4	0.5	0.0	0.0
	PrecStd	8.3	0.6	1.0	1.3	1.9	1.6	3.2	2.6	2.9	3.2	6.2	4.1	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	87.5	89.9	91.5	95.6	95.9	94.5	92.4	91.7	91.6	88.9	86.1
		MCWB	72.3	73.7	76.3	81.3	80.9	80.8	78.8	78.7	80.5	77.8	73.0	72.6
	2%	DB	83.2	85.9	88.2	90.0	92.7	93.1	90.9	89.8	90.0	90.0	87.4	84.5
		MCWB	72.2	74.6	77.6	80.8	82.2	81.6	80.3	80.2	80.9	78.7	74.7	72.0
	5%	DB	82.2	84.7	87.4	89.1	91.6	91.6	89.3	88.6	89.0	89.0	86.3	83.3
		MCWB	71.7	74.3	77.8	80.5	82.7	81.9	80.4	80.3	80.9	78.7	74.6	71.3
	10%	DB	81.1	83.7	86.6	88.3	90.8	90.4	88.0	87.4	87.9	87.9	85.1	82.0
		MCWB	71.0	73.9	77.5	80.2	82.5	81.7	80.2	80.2	80.6	78.6	74.2	70.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.3	78.8	80.9	83.5	85.7	85.4	83.6	82.8	83.3	82.6	79.9	76.9
		MCDB	81.2	84.6	86.8	89.1	90.8	91.2	89.1	87.7	88.4	88.0	85.1	82.9
	2%	WB	75.2	77.4	80.0	82.5	85.1	84.5	82.6	82.2	82.5	81.6	78.7	75.4
		MCDB	80.6	83.2	86.0	88.2	90.3	90.2	88.2	87.3	87.9	87.3	84.0	81.2
	5%	WB	74.0	76.5	79.3	81.8	84.4	83.6	81.8	81.6	81.8	80.8	77.7	74.4
		MCDB	79.7	82.4	85.3	87.6	89.7	89.4	87.3	86.7	87.2	86.7	83.2	80.2
	10%	WB	73.0	75.5	78.8	81.1	83.7	82.8	81.0	80.9	81.1	80.0	76.9	73.4
		MCDB	78.9	81.6	84.8	87.0	89.2	88.6	86.4	86.0	86.5	85.7	82.4	79.3
Mean Daily Temperature Range	5% DB	MDBR	11.1	10.9	9.3	7.7	7.5	7.0	6.7	7.1	7.9	8.9	10.4	11.3
		MCDBR	11.7	11.4	9.7	8.1	8.2	8.6	8.6	8.8	9.5	10.5	11.3	11.9
		MCWBR	5.2	5.8	5.1	4.9	6.3	5.9	5.0	4.9	5.0	4.7	5.0	5.4
	5% WB	MCDBR	9.9	9.7	8.6	7.2	7.3	7.2	7.3	7.6	8.3	8.8	9.1	10.0
		MCWBR	5.0	5.3	4.8	5.0	6.5	6.2	5.3	5.2	5.3	4.9	4.6	5.2
Clear-Sky Solar Irradiance	taub	0.512	0.507	0.581	0.586	0.635	0.635	0.567	0.545	0.549	0.544	0.518	0.534	
	taud	1.984	2.030	1.874	1.918	1.815	1.823	2.005	2.071	2.057	2.046	2.036	1.946	
	Ebn at Noon	243	254	239	239	224	222	238	245	244	240	240	232	
	Edh at Noon	54	54	65	63	69	68	56	53	54	53	51	55	
All-Sky Solar Radiation	RadAvg	1490	1770	1946	2088	1936	1446	1329	1366	1417	1470	1423	1358	
	RadStd	62	71	84	74	81	140	156	121	112	149	121	88	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	+0.54	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.54	N/A	N/A	+0.68	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page